Daily Drilling Report
SPD19B-03
4/11/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
17.0	16"	38.00	72.83	31.70	97.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	204,530.15	11,052.80	3,939,550.20		14.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
1,184.00	1,358.00		174.00	18.18	9.6

Operations at Report Time
Drilling @ 1380m.

Operations Summary
Directional drilling 16" hole section f/ 1184m to 1358m.

Operations next 24hrs
Cont' drilling 16" hole to TD. Sweep hole. Perfrom wiper trip.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/29/2014
BOP Test	4/7/2014	4/29/2014
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	3/9/2014	4/29/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	3/22/2014	5/22/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill	4/9/2014	5/25/2014
Trip Drill	4/8/2014	5/29/2014

STOP Cards

Number of Reports: 37

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00	24....	16"	D	03	P	Directional drilling 16" hole section in rotation and slide mode f/ 1184m to 1358m w/ 800 gpm/ 1950-2100 psi/ 15-25 kips/ 70-60 rpm/ 10-20 kft-lbf. Rotation Intervals Slide Intervals 1184m - 1274m 1274m - 1283m 1283m - 1333m 1333m - 1343m 1343m - 1358m ** Drilling with NL / NG. ** B/R and R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when stand down. ** Take survey before connection. ** Max gas @ 1189m = 5.8 %
00:00	00:00						Apr 12 , 2014 00:00 - 06:00 Cont' directional drilling 16'' hole section in rotation and slide mode f/ 1358m to 1380m w/ 760-1000 gpm/ 1650-2500 psi/ 5-25 kips/ 50-60 rpm/ 10-20 kft-lbf. Rotation Intervals Slide Intervals 1358m - 1360m 1360m - 1367m 1367m - 1380m ** Drilling with NL / NG. ** B/R and R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when stand down. ** Take survey before connection. >> Drilling practice in cherty zone (Expected 1371m - 1384m): High gpm up to 1000, WOB < 10 kips, Pump 20 bbl Lo-vis + 20 bbl Hi-vis every 5m drilled. >> Receive small traces of chert f/ 1372m in samples. >> Last Sample @ 1372m consists of : 100 % Lst with some dolomite and traces of chert.

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
1,358.00	Polymer Mud	9.30	48	15.0	17.000	6.0
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.7						
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	5	0.0	93.0	7.0	27,000.000	
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
3.000	5.000		0/100	48.0	1568.0	1821.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
1650.0	84.0	0.0	0.53		6.193638	0.555

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.80	7.	360	15.0	1,008.0	27.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Lafore-900	Supply Boat	4/11/2014 12:35	4/11/2014 14:40	Rec cargo f/ Morshed for Sahar 1, B/L Cargo for Kish from DD6. Dep to Sahar 1.
Morshed	Supply Boat	4/11/2014 12:24	4/11/2014 18:45	Delivered all DD6 Cargo, FW-73 MT, Cement-18 MT, C/Off & Dep for Sahar 2.
Osam Condor	Supply Boat	4/10/2014 17:05		Stdby anchorage.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bio-Cide	2.0	274.0	107.0
Bio-XCD	31.0	454.0	158.0
AMYLOSE B	46.0	800.0	434.0
H2S Scavenger	5.0	120.0	53.0
Defoam SLC	5.0	220.0	69.0
Pac-R	12.0	360.0	126.0
Pac-UL	15.0	230.0	95.0
Caustic Soda	19.0	825.0	286.0
Soda Ash		312.0	101.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	15.0
Barite Bulk	0.0	0.0	147.0
Cement G	0.0	18.0	168.0
Bentonite Bulk	0.0	0.0	65.0
Bentonite BB	0.0	0.0	21.0
Diesel Fuel	12.0	0.0	140.0
Potable Water	20.0	20.0	110.0
Drill Water	120.0	73.0	573.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
5	16	Varel	VM516DGXU	6004481	PDC	M322	13/13/13/13/13/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.04	43.08	418,282.80	878.00	1,358.00	480.00	11.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	890	1,650.0	2,100.0	10	25	104	181
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
54.90	37.21	86.37	13.63	343	301	237	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
1,184.00	1,274.00	8.10	17,000.0	800	70	2,050.0	18	301
1,274.00	1,283.00	3.30		800		2,000.0	18	301
1,283.00	1,333.00	3.98	19,000.0	800	70	2,000.0	15	301
1,333.00	1,343.00	1.72		800		1,950.0	10	301
1,343.00	1,358.00	1.08	20,000.0	800	60	2,000.0	25	301

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
15 jts 5 7/8" HWDP	5 7/8	4.250	139.60	313.30
XOS	8	3.000	0.79	173.70
6 jts 8 1/4" DC	8 1/4	2.810	56.24	172.91
8" DA Drilling Jar (COR2898AD)	8	3.500	5.72	116.67
8 jts 8 1/4" DC	8 1/4	2.810	75.34	110.95
XOS	9 1/2	3.000	1.22	35.61
9 1/2" Float sub with valve	9 1/2	3.000	0.78	34.39
9 1/2" NMDC	9 1/2	4.000	9.44	33.61
9 1/2" TM-Sub (Mwd and gamma probes inside)	9 1/2		2.11	24.17
9 1/2" NMDC	9 1/2	4.000	9.44	22.06
9 1/2" Restrictor Sub (Jet: 36)	9 1/2	4.300	0.92	12.62
14 3/4" String Stabilizer (COR3032AD)	14 3/4	3.000	2.16	11.70
9 5/8" Motor (BH: 1.25 , Sleeve: 15 3/4")	9 5/8		9.14	9.54

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
16" PDC Bit (SN: 6004481)	16		0.40	0.40

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
1	1,358.00	14.00	30	170.0	97	170
1	1,358.00	14.00	40	250.0	97	227
1	1,358.00	14.00	20	115.0	97	114
2	1,358.00	14.00	20	125.0	97	114
2	1,358.00	14.00	30	170.0	97	170
2	1,358.00	14.00	40	245.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)
400.0	8.0	400.00		11.48	1,650.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
1,193.36	244.20	22.10	1,142.00	-138.69	-265.13	299.21	1.05	-1.03
1,222.12	244.10	21.50	1,168.71	-143.34	-274.74	309.88	0.63	-0.63
1,251.33	243.60	21.10	1,195.92	-148.02	-284.26	320.49	0.45	-0.41
1,279.81	245.10	20.50	1,222.55	-152.40	-293.38	330.60	0.85	-0.63
1,308.93	245.10	19.70	1,249.89	-156.61	-302.46	340.60	0.82	-0.82
1,338.09	244.90	18.80	1,277.42	-160.67	-311.17	350.20	0.93	-0.93

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Lower Dariyan		1,183.50	1,132.89
Upper Gadvan		1,209.00	1,156.52
Khalij		1,231.00	1,177.01
Lower Gadvan		1,247.00	1,191.88
Fahliyan		1,286.50	1,228.82

General Remarks

Comment

>>> Requirements: NIL

>>> Sent:

*** Lafore-900 boat to Kish base: (4 jts damaged 13 3/8" casing)

>>> Received:

*** Morshed boat f/ Kish base: (AL-Ghaith 13 3/8" CSG handling tools: 1 no Power unit + 1 no FMS slips without 13 3/8" Jaws + 1 no control panel for FMS) + (PDF chemicals: 2 BB Soda ash + 5 pallet Caustic soda + 2 pallet MI Cide + 1 pallet Pac-R + 2 pallet Pac-UL + 1 pallet Bio-XCD + 10 pallet Starch)